Information Technology Institute

ITI

Introduction to Computer Networks & Cyber Security

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Cyber Security Essentials

Part 2 (Cyber Security Essentials)

- **Session Outlines**
  - **Information Security Goals**
    - Confidentiality ,Integrity, Availability
  - **Risks & Threats**
    - Threats & Vulnerabilities
    - Attackers methodology & Methods
    - Malware Types
  - **Security Defenses**
    - Firewalls (Static & Dynamic firewalls)
    - IDS /IPS
    - VPN
    - Proxy
    - Next generation Firewalls
  - **Encryption**
    - Symmetric & Asymmetric Key Cryptography

Cyber Security Essentials

- People use networks to exchange sensitive information with each other.
- People purchase products and do their banking over the Internet.
    - We rely on networks to be secure and to protect our identities and our private information
- Cyber Security is a shared responsibility that each person must accept when they connect to the network.



Cyber Security Essentials

- **Cyber Security**
  - **How to protect systems, networks, programs, devices and data from any attacks**

- **Cyber Security involves protecting resources:**
  - **End-user resources**: PCs, Laptop, Tablets
  - **Network resources**: Routers, Switches
  - **Server resources**: Rack Mount, Blade servers



Cyber Security Essentials

Security Goals

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Security Goals

- **Confidentiality**
  - Ensuring that information is not revealed to unauthorized persons
  - Data transmitted or stored should only be revealed to an intended audience
- **Integrity**
  - Ensuring **consistency** of data
  - It should be possible to detect any modification of data
- **Availability**
  - Ensuring that legitimate users are not denied access to information and resources

Focus of Security is Risk

- Security deals with managing risk to your critical assets

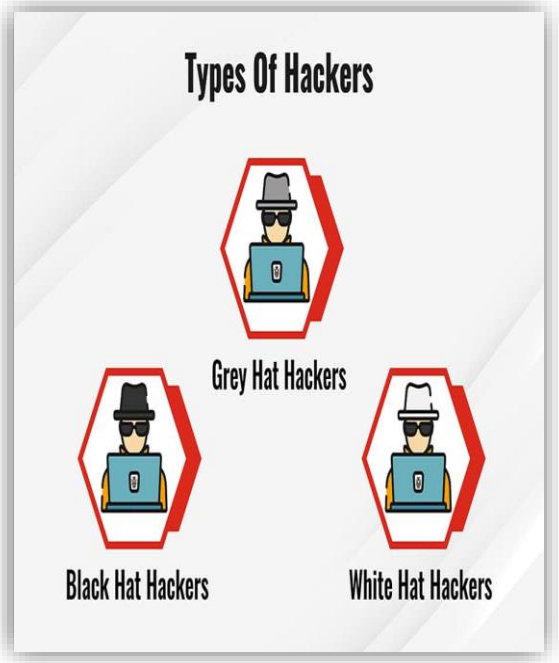
Risk = Threat x Vulnerabilities x Impact

- It's **impossible** to totally eliminate risk
- Security 99.9 % Not found Why ?
  - This can be seen through the different types of attacks that users face today.
  - New technologies / applications
  - New Vulnerabilities
  - the difficulties in defending against these attacks

Vulnerability is the degree of weakness which is found in every network and device.
Threats is A person, thing, event or idea which poses danger to **an asset** in terms of that asset's confidentiality, integrity, availability or legitimate use

Attackers Types

- **Black hats (Offensively)**
  - These are cybercriminals who exploit system vulnerabilities for personal gain, financial profit, or to cause damage.
  - Known as '**Crackers**.'
- **White Hats (Defensively)**
  - they are authorized professionals who test systems to identify and fix security flaws. They work legally for organizations or governments to strengthen cybersecurity
  - Known as '**Security Analysts**, **Ethical hacker**'.
- **Gray Hats (Offensively) (Defensively)**
  - Operate without permission but usually without malicious intent. They may expose vulnerabilities publicly or report them to organizations



What does a Malicious Hacker Do?

- Reconnaissance
- Scanning
- Gaining access
- Maintaining access
- Covering tracks



Reconnaissance (Phase 1)

- Reconnaissance refers to the preparatory phase where an attacker seeks to gather as much information as possible about a target of evaluation prior to launching an attack.
- Gathering info about internal structure of organization, by browsing and search the internet









Scanning (Phase 2)

- Scanning refers to pre-attack phase when the hacker scans the network with specific information gathered during reconnaissance.
- Scanning for open ports, operating systems, applications, open shares,

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- In your lab use A port scanner tools to find the open ports on your device

Part 2 _Gaining Access (Phase 3)

- Gaining Access refers to the true attack phase. The hacker exploits the system.
- The exploit can occur over a LAN, locally, Internet.
- Examples include buffer overflows, denial of service, session hijacking etc.

Maintaining Access (Phase 4)

- Maintaining Access refers to the phase when the hacker tries to retain his ‘ownership’ of the system.
- Sometimes, hackers harden the system from other hackers as well (to own the system).

Covering Tracks(Phase 5)

- Covering Tracks refers to the activities undertaken by the hacker to extend his misuse of the system without being detected.

- Reasons include need for continued use of resources, removing evidence of hacking, avoiding legal action etc.

- Hackers can remain undetected for long periods.

Cyber Attacks

Cyber Attack is any attempt to destroy, expose, alter, disable, steal or gain unauthorized access to or make unauthorized use of an asset



Attack Types

- Passive Attack
- Active Attack
- Phishing Attack
- Hijack Attack
- Spoof Attack
- Buffer Overflow Attack
- Exploit Attack
- Password Attack



Passive attack VS Active attack

- **Passive attack** attempts **to take the information** from the system and **does not affect any system resources and its operations.**
  - **Ex : Cookies , Spyware , Wireshark**

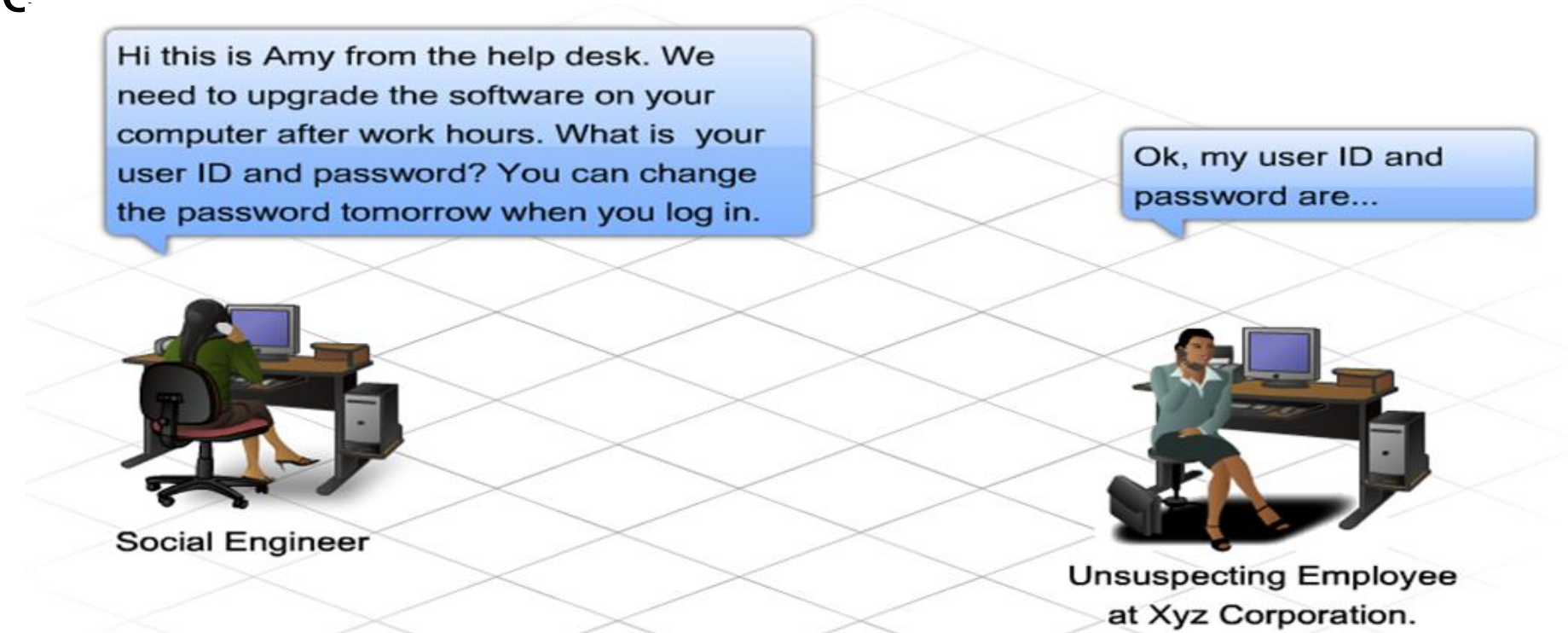


- **Active attack** attempts to **change** the system resources or affect their usual operations.
  - **Ex : Ransomware, Viruses, worms**



Social engineering

- **Social engineering** is a term that refers to the ability of something or someone to influence the behavior of a group of people



PHISHING ATTACK

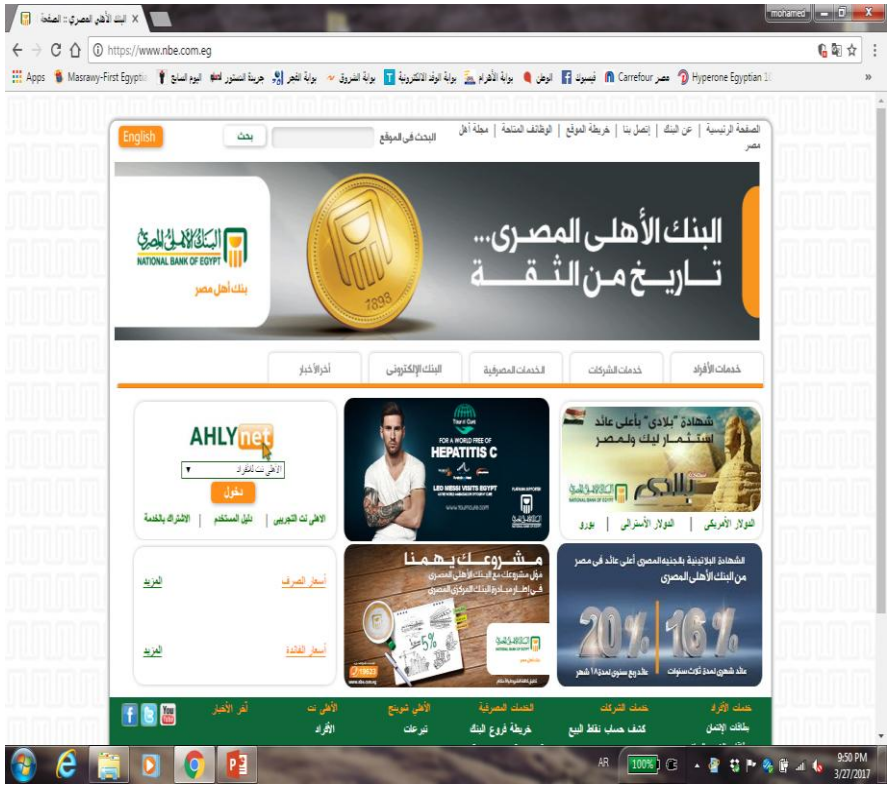
In **phishing attack**

- the hacker creates a fake web site that looks exactly like a popular site.
- **The phishing part of the attack is that the hacker sends**
  - **An e-mail message , Sms message**
- **trying to trick the user into clicking a link that leads to the fake site.**
  - When the user attempts to log on with their account information, the hacker records the username and password and then tries that information on the real site.



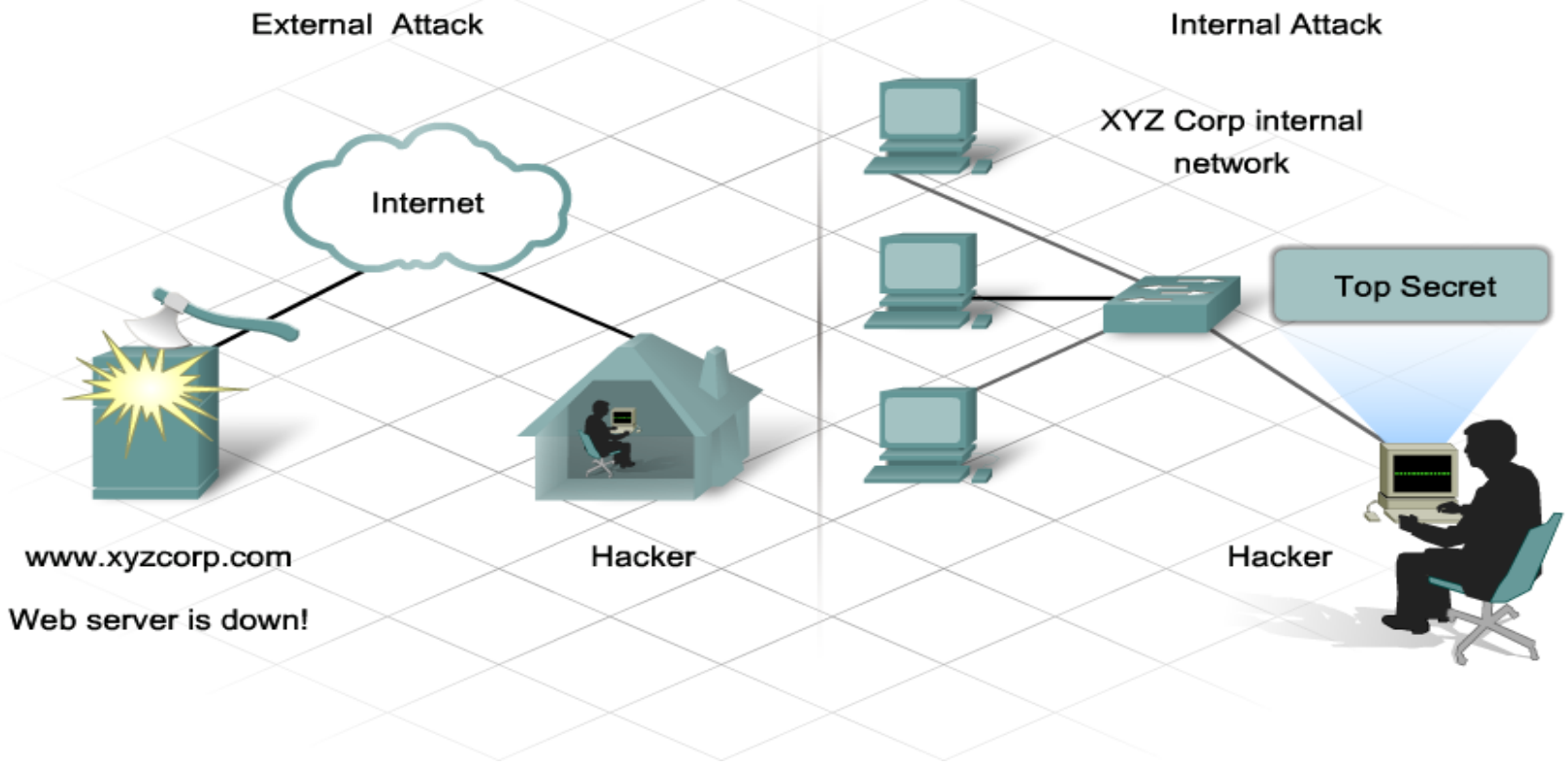
Part 2_Find the fake one ?





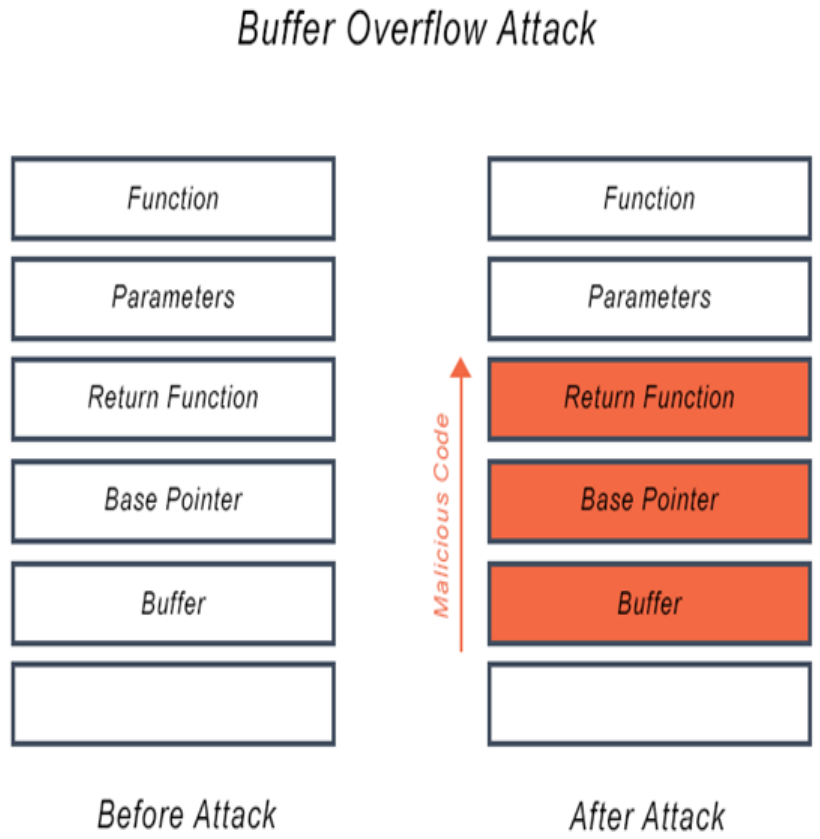
https://www.virustotal.com/gui/home/url

Insider Attack



Part 2_BUFFER OVERFLOW ATTACK

A **buffer overflow** attack is when the attacker sends more data to an application than is expected. A buffer overflow attack usually results in the attacker gaining administrative access to the system in a command prompt or shell.



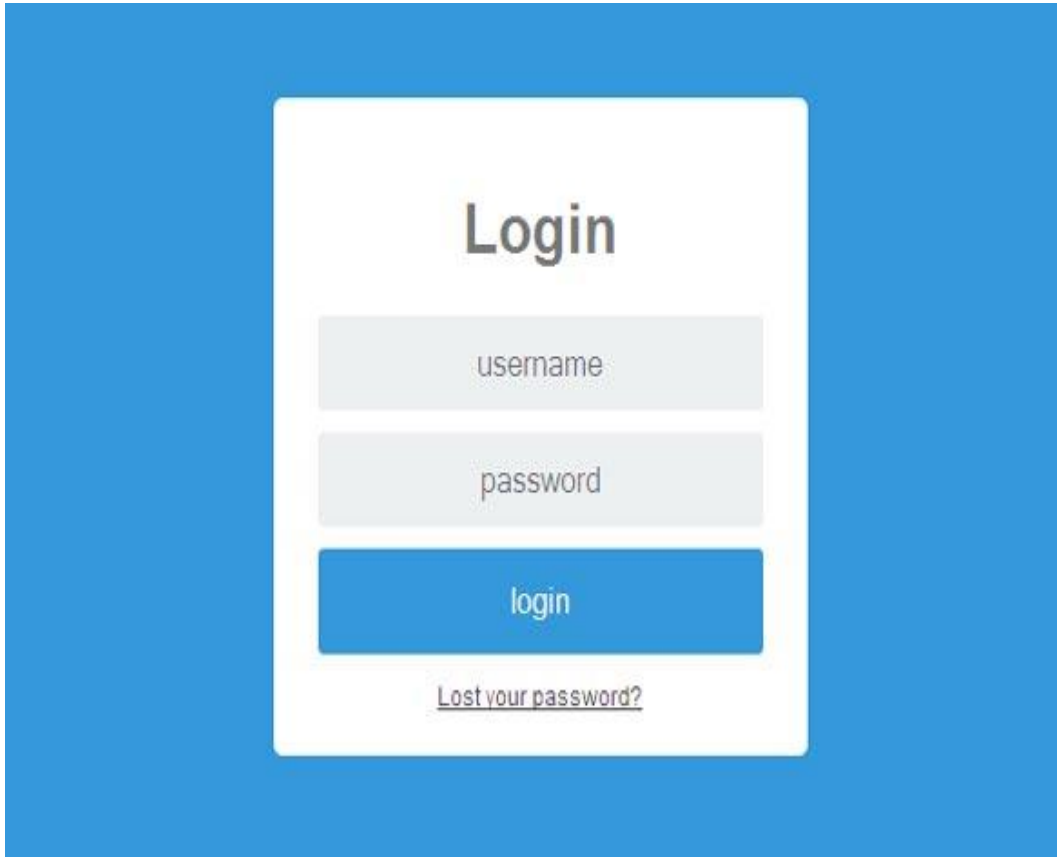
Part 2_PASSWORD ATTACK

An attacker tries to crack the passwords stored in a network account database or a password-protected file.



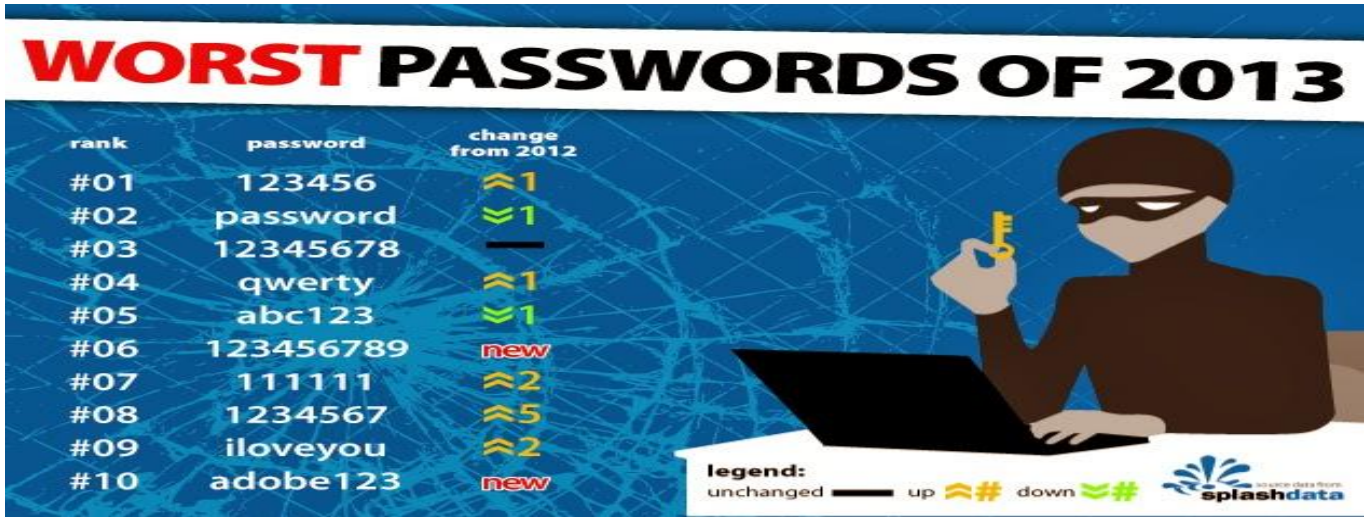
Part 2_types of password attack

- Dictionary attack
- Brute-force attack
- Hybrid attack.



Part 2_types of password attack

- A **dictionary attack** uses a word list file, which is a list of potential passwords.



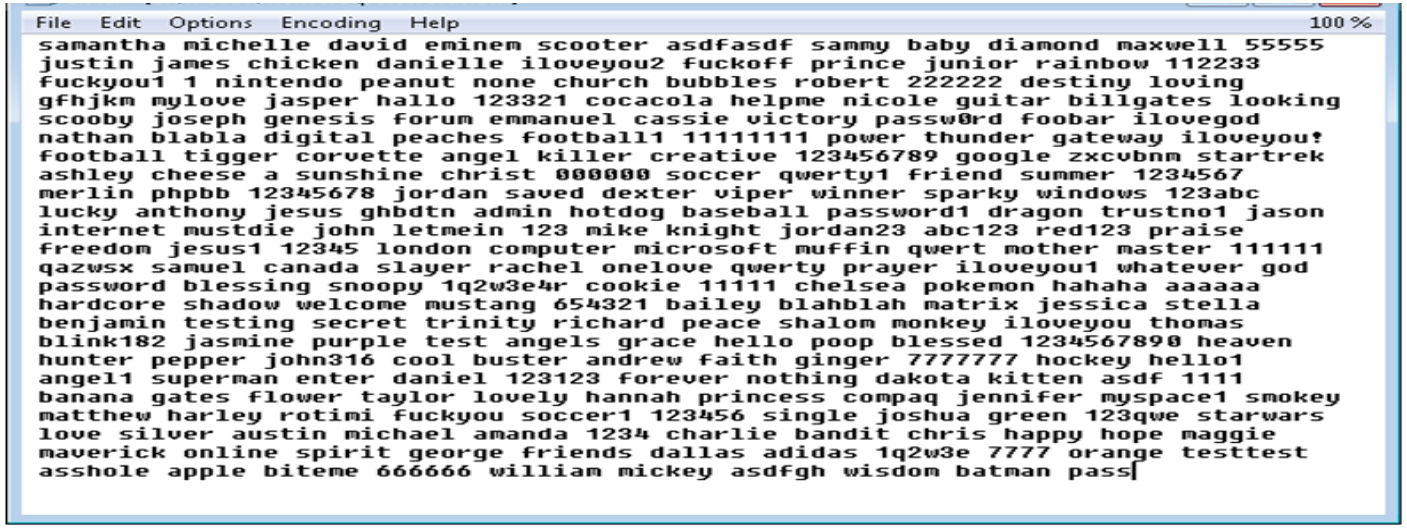
Part 2_TYPES OF PASSWORD ATTACK

- A **brute-force attack** is when the attacker tries every possible combination of characters.



Part 2_types of password attack

- A **hybrid attack** builds on the dictionary attack method by adding numerals and symbols to dictionary words

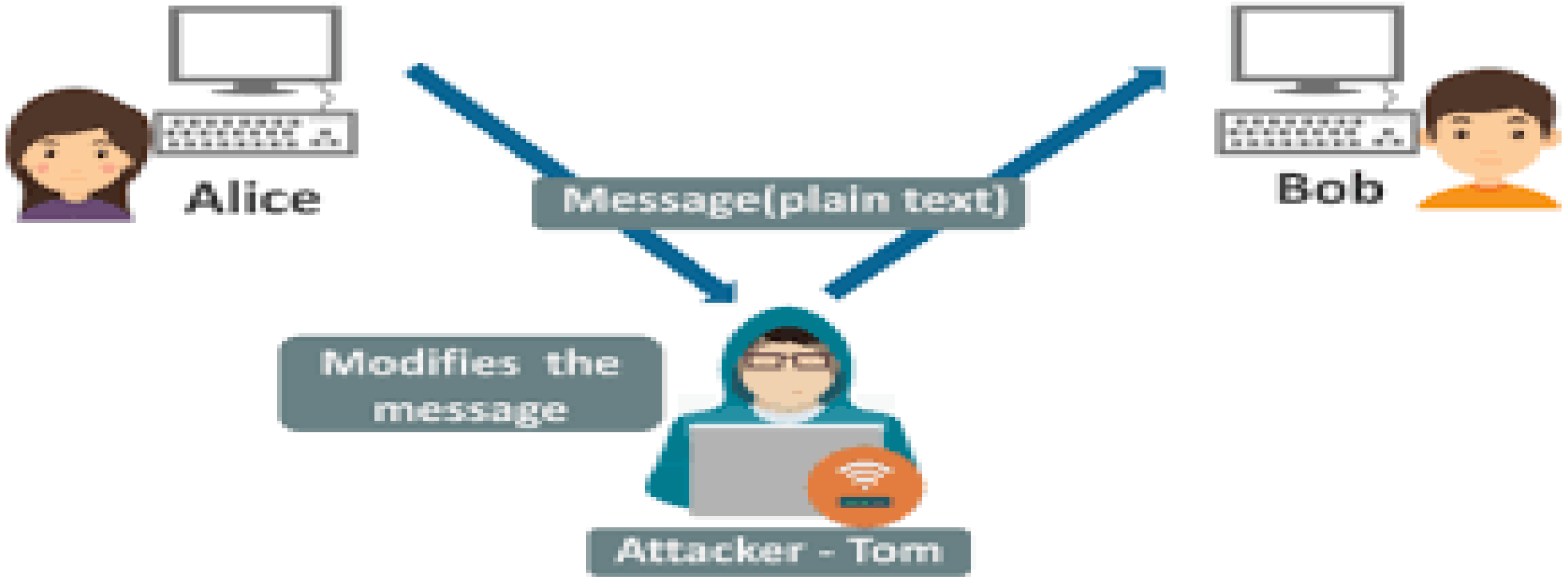


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- Check you compromised password and your data leakage at

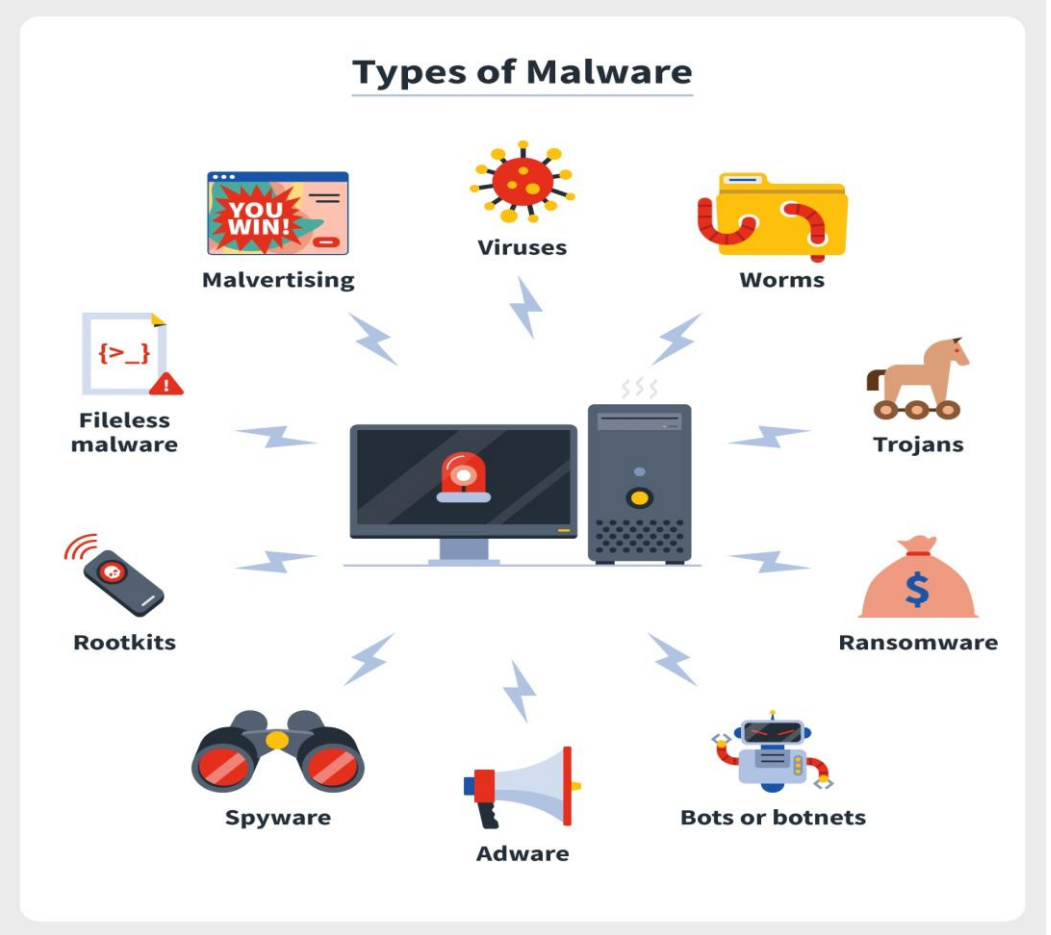
  - **https://haveibeenpwned.com/**

Man in the middle Attack

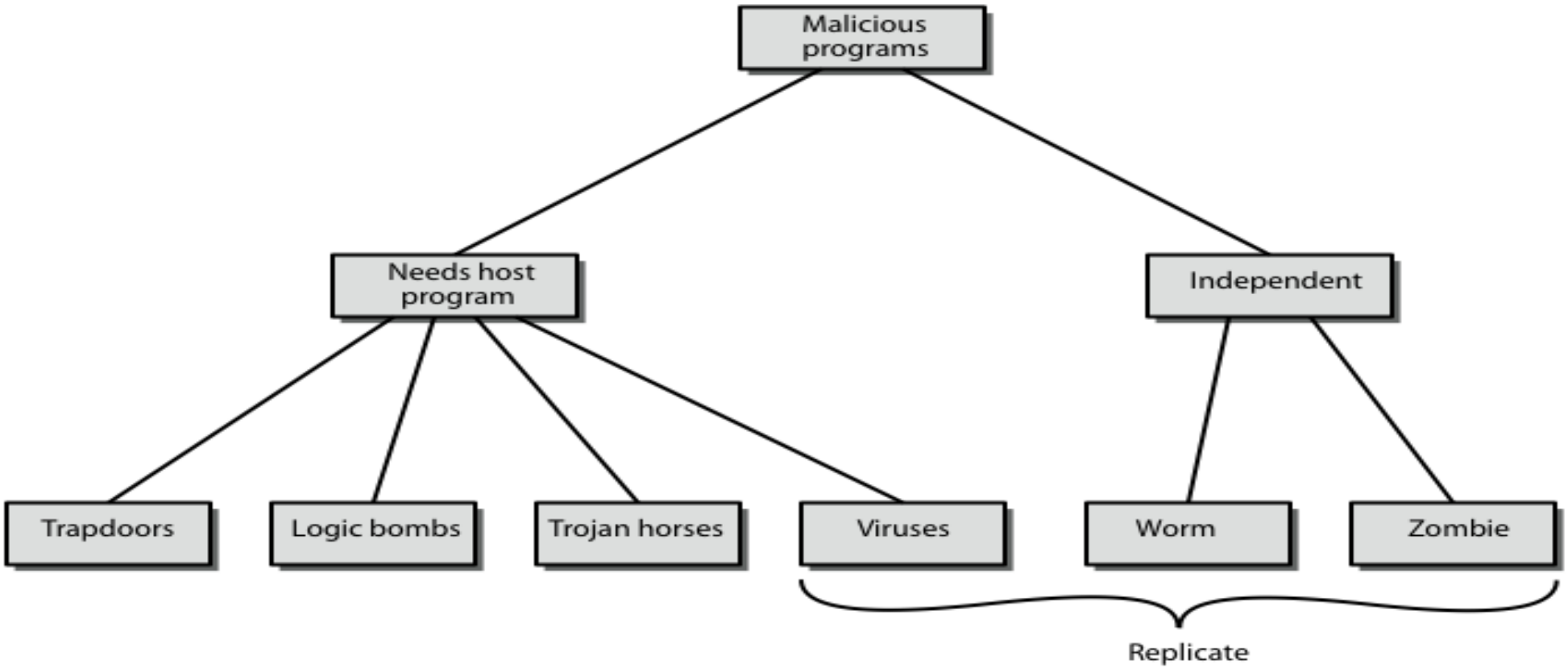


Part 2_Malware Types

- Malware Capabilities
  - Destruction of Data
  - Leaking Confidential Information
  - Providing Backdoor Access
  - Countless Other Opportunities

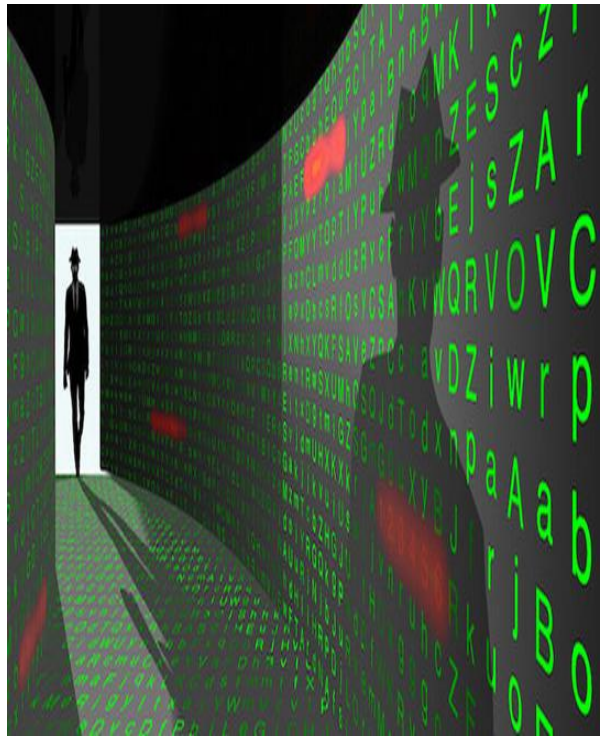


Malicious Software



Backdoor or Trapdoor

- Secret entry point into a program
- Allows those who know access bypassing usual security procedures
- Have been commonly used by developers
- Requires good s/w development & update
- Can’t be removed or scanned and the only way is to uninstall sw or format the system



Viruses

- A virus is malicious software that is attached to another program to execute a particular unwanted function on a user's workstation.

- Both propagates itself & carries a payload
  - Carries code to make copies of itself
  - As well as code to perform some covert task



Trojan Horse

- program with hidden side-effects
- which is usually superficially attractive
  - eg game, software upgrade etc
- when run performs some additional tasks
  - allows attacker to indirectly gain access they do not have directly
- often used to propagate a virus/worm or install a backdoor or simply to destroy data
- Open some ports or pass some malicious files



Independent malware

- Worms
- Zombie
- Man in the middle
- DOS
- DDOS
- Spyware and Tracking Cookies

Worms

- Replicating but **not infecting** program
- Typically spreads over a network
- Using users distributed privileges or by exploiting system vulnerabilities
- Widely used by hackers to create **zombie pc's**, subsequently used for further attacks, especially dos
- Major issue is lack of security of permanently connected systems



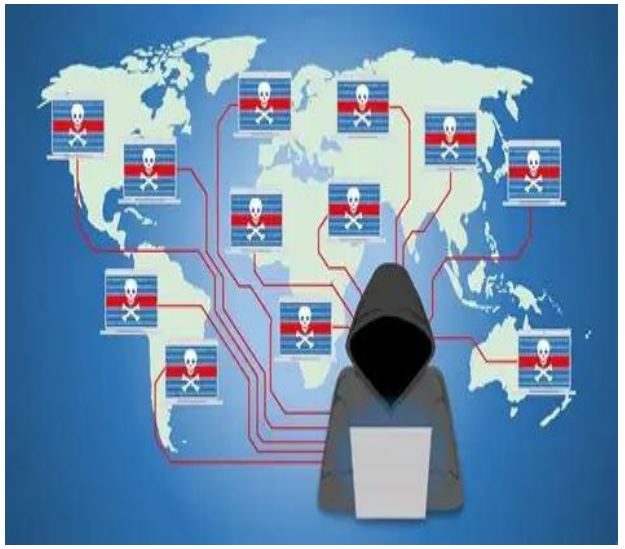
Lab

- Check you downloaded software at
  - https://www.virustotal.com/gui/home/upload

Zombie (Botnet)

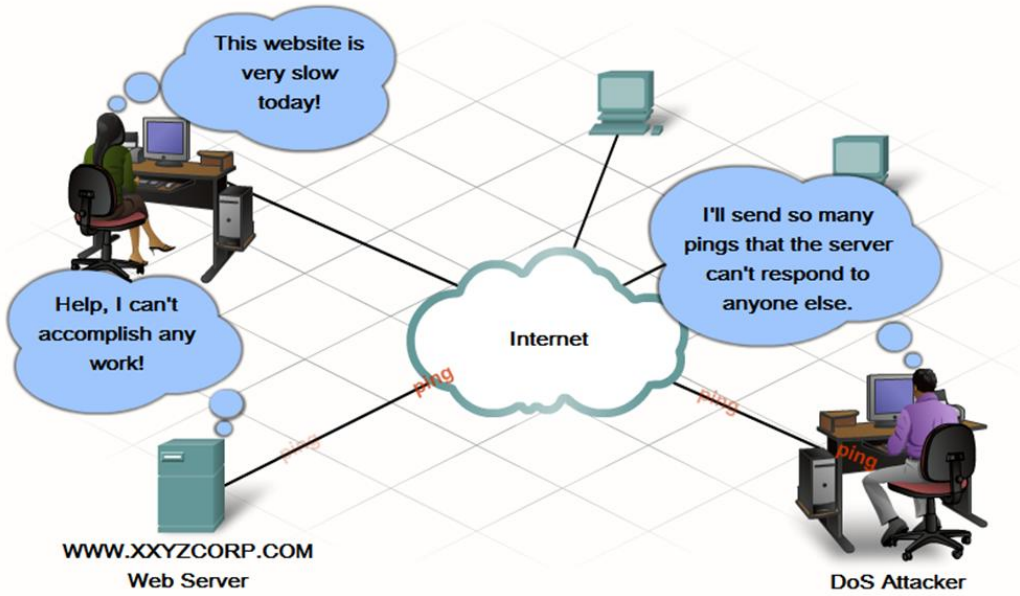
- Program which secretly takes over another networked computer then uses it to indirectly launch attacks
- Often used to launch distributed denial of service (DDoS) attacks





DoS Attack

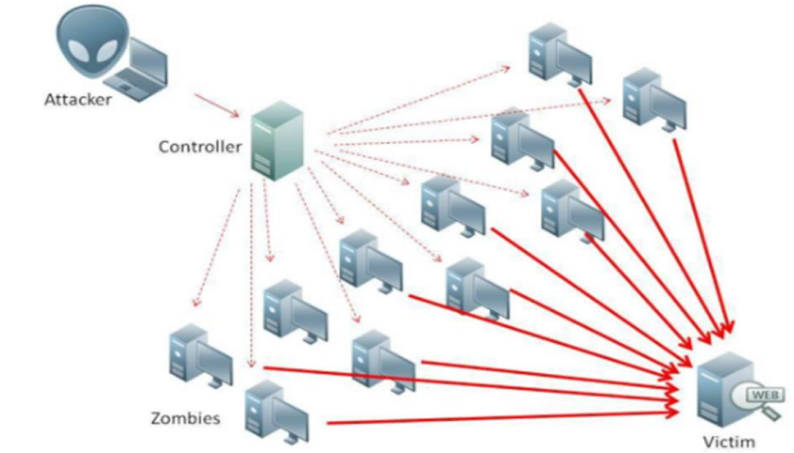
- Denial of service is about without permission knocking off services, for example through crashing the whole system.
- This kind of attacks are easy to launch and it is hard to protect a system against them.
- Consume host resources
  - Memory
  - Processor cycles
- Consume network resources
  - Bandwidth

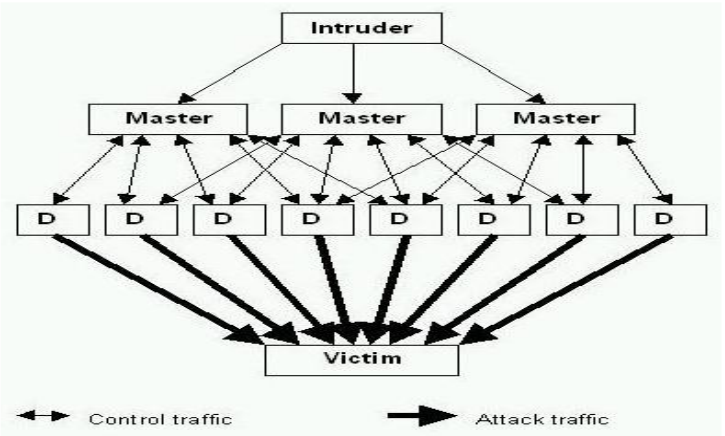


DDoS Attack

- DDoS – A distributed denial of service attack uses multiple machines to prevent the legitimate use of a service.

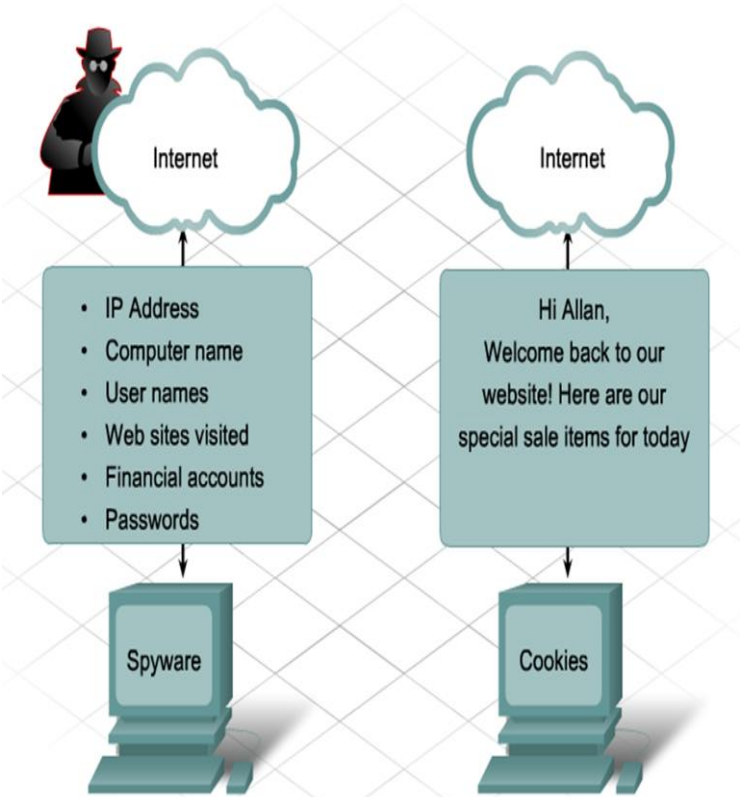
- Making networked systems unavailable by flooding with useless traffic using large numbers of "zombies" growing sophistication of attacks





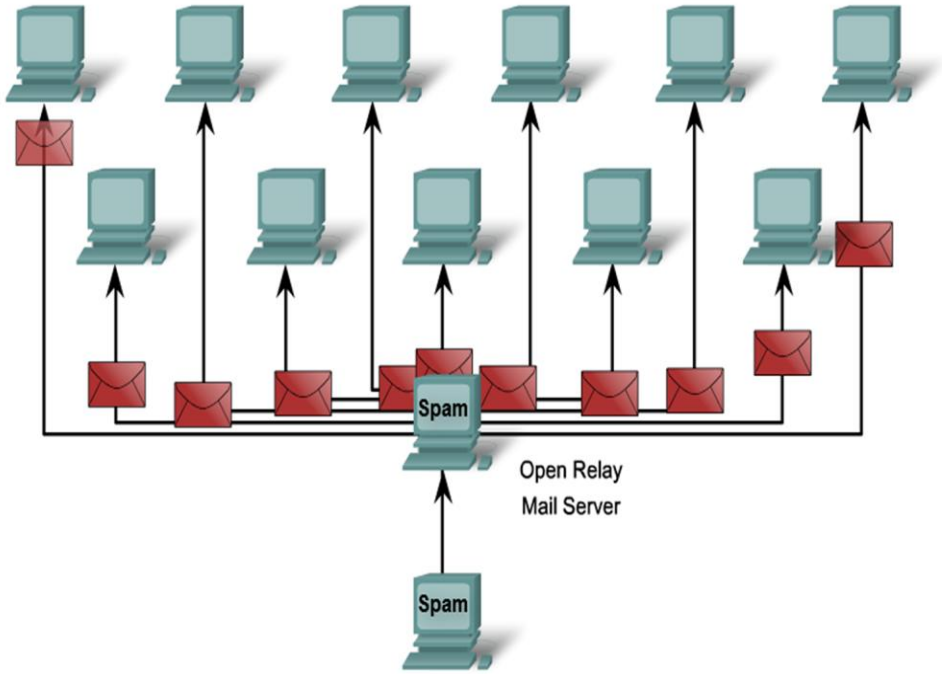
Spyware and Tracking Cookies

- **Spyware**
  - Spyware is any program that gathers personal information from your computer **without your permission or knowledge**. This information is **sent to advertisers or others** on the Internet and can include passwords and account numbers.
- **Tracking Cookies**
  - Cookies are a **form of spyware** but are **not always bad.** They are used to record information about an Internet user when they visit websites.



Spams

- **Spams**
- is a serious network threat that can overload ISPs, email servers and individual end-user systems.
- A person or organization responsible for sending spam is called a spammer.
- Spammers often make use of unsecured email servers to forward email.
- Spammers can use hacking techniques, such as viruses, worms and Trojan horses to take control of home computers.



Thank You